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Secure Authentication and Authorization Framework for Healthcare IoT Systems

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by

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Declaration of Academic Honesty and Integrity

We declare that this project report entitled "MEDISECURE: Secure Authentication and Authorization Framework for Healthcare IoT Systems" is the result of our original work carried out during the Summer Internship 2026 at the Centre of Excellence for Digital Manufacturing and Engineering (CoE-DMA), National Institute of Technology Warangal.

We certify that this report has not been submitted, either in part or in full, for the award of any other degree, diploma, or certificate at any institution.

Wherever the work, ideas, or findings of other researchers have been referred to in this report, proper acknowledgment has been made through suitable citations and references. We have adhered to the principles of academic honesty and integrity and have neither fabricated nor falsified any information presented in this report.

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ABSTRACT

Healthcare systems require secure patient identification, efficient medical record management, and real-time health monitoring to improve the quality of patient care. Traditional methods often rely on manual record keeping and weak authentication mechanisms, leading to delays, security issues, and data management challenges. To address these problems, this project presents MEDISECURE: Secure Authentication and Authorization Framework for Healthcare IoT Systems.

The proposed system integrates biometric authentication, IoT devices, and full-stack web technologies to provide a secure and efficient healthcare management platform. An Arduino UNO R4 WiFi is used to collect patient health data from the Pulse Sensor and AD8232 ECG Sensor, while a R307/AS608 Fingerprint Sensor provides secure patient authentication. The system is developed using React.js for the frontend, Node.js with Express.js for the backend, and MySQL for centralized storage of patient records, prescriptions, and sensor data. Communication between the hardware and backend is established using SerialPort, enabling real-time transmission of sensor readings.

The developed application provides separate dashboards for patients, doctors, and hospital staff, allowing secure login, patient registration, medical record management, live health monitoring, and prescription updates. The system successfully demonstrates secure fingerprint-based authentication, centralized electronic health record management, and real-time monitoring of pulse rate and ECG signals.

MEDISECURE offers a reliable and scalable Healthcare IoT solution that enhances patient safety, improves healthcare efficiency, and reduces manual documentation. The proposed framework can be further extended with cloud integration, AI-based disease prediction, mobile application support, and deployment in real hospital environments.

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INTRODUCTION

1.1 Introduction

Healthcare is one of the most essential sectors where timely access to patient information and secure management of medical records directly influences the quality of treatment and patient safety. With the rapid advancement of digital technologies, hospitals are gradually transforming from conventional paper-based systems to intelligent healthcare solutions that utilize the Internet of Things (IoT), cloud computing, and embedded systems. These technologies enable continuous patient monitoring, efficient data management, and improved communication between healthcare professionals.

Traditional hospital management systems often depend on manual documentation and password-based authentication methods. Such systems are time-consuming, prone to human errors, and vulnerable to unauthorized access. Patient records are frequently scattered across multiple departments, making it difficult for doctors to retrieve accurate medical information during emergencies. These limitations increase administrative workload, delay diagnosis, and reduce the overall efficiency of healthcare services. The Internet of Things (IoT) has emerged as a promising technology for modern healthcare by connecting medical devices, sensors, and software applications through the internet. IoT-based healthcare systems allow real-time monitoring of patients, automatic collection of physiological data, and secure sharing of medical information. This improves clinical decision-making and enhances the quality of healthcare delivery.

To address these challenges, MEDISECURE: Secure Authentication and Authorization Framework for Healthcare IoT Systems has been developed. The proposed system integrates biometric authentication, embedded systems, and full-stack web technologies into a unified healthcare platform. The system utilizes an Arduino UNO R4 WiFi as the main controller, an R307/AS608 Fingerprint Sensor for secure patient authentication, a Pulse Sensor for heart rate monitoring, and an AD8232 ECG Sensor for real-time electrocardiogram acquisition. The frontend is developed using React.js, while the backend uses Node.js and Express.js. Patient records, prescriptions, and sensor readings are securely stored in a MySQL database. Communication between the hardware and backend is established using SerialPort, enabling seamless transfer of sensor data.

The developed system provides dedicated dashboards for doctors, patients, and hospital staff. It enables secure patient authentication, centralized medical record management, real-time monitoring of pulse and ECG signals, and efficient prescription management. By integrating IoT technologies with modern web applications, the proposed system contributes towards the realization of Healthcare 4.0 and supports the digital transformation of healthcare services.

1.2 Need for the Project

Modern healthcare systems require secure authentication, efficient management of patient records, and continuous monitoring of health parameters. Conventional methods of patient identification and record maintenance are often slow, error-prone, and susceptible to unauthorized access. In emergency situations, delays in retrieving patient information can affect the quality of medical treatment.

The increasing adoption of IoT technologies provides an opportunity to improve healthcare services through automation, real-time monitoring, and secure communication between medical devices and software systems. Therefore, there is a need for an integrated healthcare platform that combines biometric authentication, IoT-based health monitoring, and centralized medical record management to improve patient safety, healthcare efficiency, and data security.

1.3 Objectives of the Project

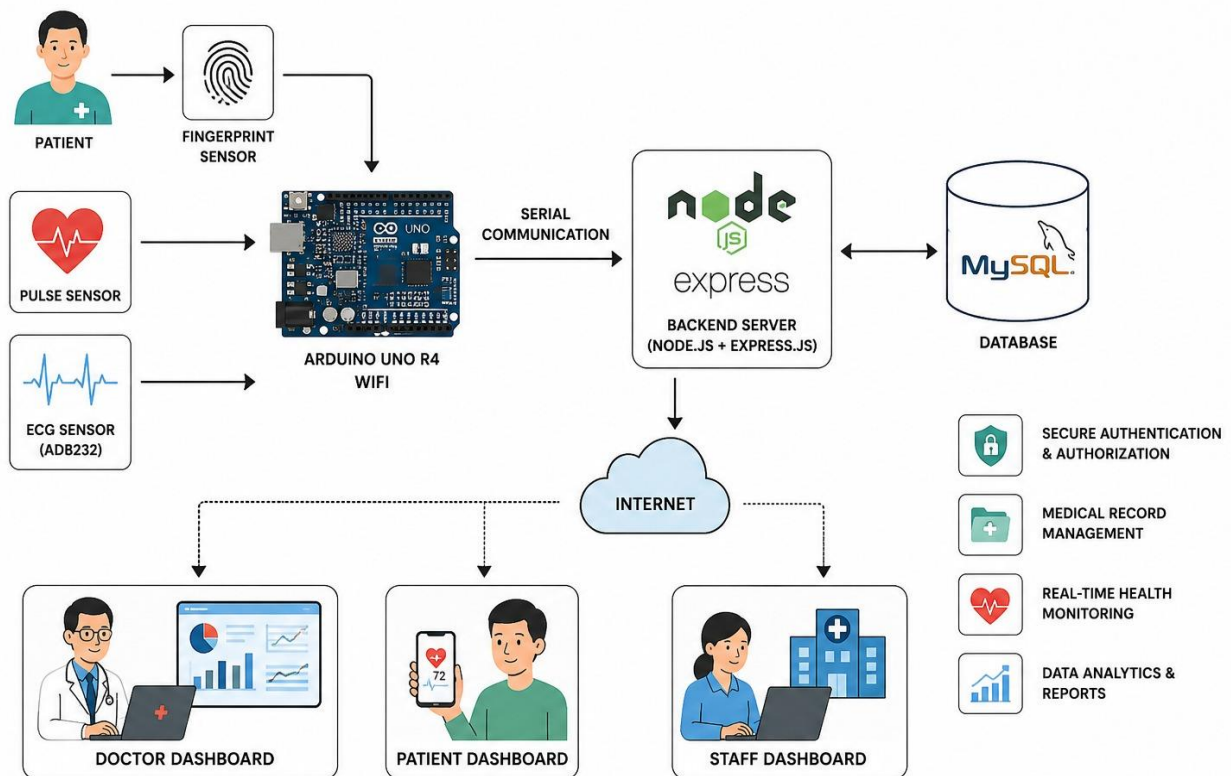
The main objectives of the proposed system are:

- To develop a secure Healthcare IoT platform for patient authentication and medical record management.
- To implement fingerprint-based biometric authentication for secure patient identification.
- To monitor patient pulse rate and ECG signals in real time using IoT sensors.
- To provide separate dashboards for doctors, patients, and hospital staff.
- To securely store patient information, prescriptions, and sensor readings in a MySQL database.
- To establish seamless communication between hardware and software using SerialPort and REST APIs.
- To reduce manual documentation and improve healthcare efficiency through digital automation.

1.4 Scope of the Project

The scope of MEDISECURE includes secure patient authentication, centralized storage of electronic medical records, and real-time monitoring of physiological parameters. The system supports patient registration, fingerprint verification, prescription management, and visualization of pulse and ECG data through web-based dashboards. The proposed framework can be further enhanced with cloud integration, mobile application support, AI-assisted disease prediction, and deployment in hospitals and healthcare centres.

OVERALL CONCEPT OF SMART HEALTHCARE IOT SYSTEM



REVIEW OF LITERATURE

2.1 Introduction

The rapid advancement of the Internet of Things (IoT), embedded systems, and cloud computing has significantly transformed the healthcare industry. Modern healthcare systems aim to provide secure patient authentication, continuous health monitoring, and efficient medical record management through interconnected smart devices. Several researchers have proposed IoT-based healthcare frameworks that integrate wearable sensors, biometric authentication, and electronic health records to improve healthcare services. This chapter reviews existing research related to Healthcare 4.0, biometric authentication, IoT-based patient monitoring, and digital healthcare management. It also highlights the research gap addressed by the proposed MEDISECURE system.

2.2 Review of Existing Literature

2.2.1 Lightweight Three-Factor Authentication Protocol for 6G-Enabled Healthcare Systems using Chebyshev Chaotic Maps and BioHashing

Abbas et al. proposed a lightweight three-factor authentication protocol designed for secure communication in 6G-enabled healthcare systems. The protocol combines three authentication factors: password, smart card, and biometric information to provide strong user authentication while maintaining low computational complexity. The authors employed Chebyshev chaotic maps for secure session key generation and BioHashing techniques to protect biometric templates. The proposed scheme eliminates the need for a secure communication channel during the registration phase and is suitable for resource-constrained healthcare IoT devices. Formal security verification demonstrated resistance against replay attacks, impersonation attacks, man-in-the-middle attacks, and password guessing attacks. Performance evaluation also showed reduced communication overhead, storage requirements, and computational cost compared to existing authentication protocols.

Although the protocol provides excellent authentication security, it mainly focuses on secure communication and user verification. It does not include real-time health monitoring, electronic medical record management, or integration with IoT-based medical sensors.

The proposed MEDISECURE system extends this concept by integrating secure fingerprint authentication with real-time Pulse and ECG monitoring using Arduino UNO R4 WiFi, React.js, Node.js, and MySQL, thereby providing a complete Healthcare IoT solution.

2.2.2 A Lightweight and Secure Mutual Authentication Scheme for Smart Healthcare Systems in Cloud Environments

Hammami et al. presented a lightweight mutual authentication protocol for cloud-enabled smart healthcare systems. The proposed scheme utilizes Elliptic Curve Cryptography (ECC) and secure hash functions to establish mutual authentication between users and healthcare servers while ensuring user anonymity and perfect forward secrecy. The protocol effectively protects against replay attacks, impersonation attacks, insider attacks, and man-in-the-middle attacks. Performance analysis indicated reduced computational complexity, communication overhead, and storage cost compared with several existing authentication methods.

However, the proposed work primarily focuses on secure authentication and cloud communication. It does not provide real-time physiological monitoring, biometric patient identification, or integrated hospital management through dedicated dashboards.

The MEDISECURE project overcomes these limitations by integrating fingerprint authentication with real-time Pulse and ECG monitoring, centralized patient record management, and web-based dashboards for doctors, patients, and hospital staff.

2.2.3 Advances in Fog Computing and the Internet of Things for Smart Healthcare

This work discusses the application of Fog Computing and the Internet of Things (IoT) in modern healthcare systems. It explains how fog computing processes medical data closer to IoT devices, thereby reducing latency and improving real-time patient monitoring. The book covers healthcare architectures, wearable sensors, electronic health records, artificial intelligence, blockchain technology, and data security. It emphasizes the importance of decentralized processing for improving healthcare efficiency and reducing dependence on cloud servers.

Although the book provides a comprehensive overview of Healthcare 4.0 technologies, it mainly focuses on theoretical concepts and emerging architectures rather than a practical implementation of a secure healthcare management system.

The proposed MEDISECURE system implements these concepts in practice by integrating biometric authentication, IoT sensors, embedded hardware, and full-stack web technologies into a unified healthcare platform.

2.2.4 Fundamentals of Fog Computing and the Internet of Things for Smart Healthcare

This book presents the fundamental concepts of fog computing and IoT-based healthcare systems. It discusses the architecture of smart healthcare, networking protocols, security challenges, healthcare data management, and interoperability among medical devices. The authors explain how edge computing improves response time and reduces communication delays for healthcare applications. Future technologies such as artificial intelligence and blockchain are also discussed as potential enhancements for digital healthcare.

However, the book mainly provides theoretical knowledge and generalized healthcare architectures without implementing a complete hospital management system integrating biometric authentication, physiological monitoring, and web-based dashboards.

The MEDISECURE system bridges this gap by implementing a complete Healthcare IoT framework that combines fingerprint authentication, Pulse and ECG monitoring, electronic medical record management, and secure communication between hardware and software components.

2.3 Research Gap

The literature survey indicates that considerable research has been carried out in the areas of secure authentication, IoT-based healthcare monitoring, cloud computing, and electronic health record management. Existing studies have successfully addressed individual challenges such as secure user authentication, cloud-based communication, remote patient monitoring, and Healthcare 4.0 technologies. However, most of these solutions focus on only one or two aspects of healthcare and do not provide a unified platform that integrates biometric authentication, real-time health monitoring, and centralized medical record management.

Many authentication schemes emphasize secure communication between healthcare users and servers but lack integration with physiological sensors for continuous patient monitoring. Similarly, IoT-based healthcare systems provide real-time monitoring of vital signs but often rely on traditional authentication mechanisms, making them vulnerable to unauthorized access. Furthermore, several healthcare architectures discussed in the literature are theoretical or cloud-centric and do not demonstrate the practical implementation of embedded hardware integrated with modern web technologies.

Another important limitation observed is the absence of dedicated interfaces for different healthcare users such as patients, doctors, and hospital staff. Most existing systems also do not provide seamless communication between IoT devices, backend servers, and databases for efficient storage and retrieval of electronic medical records.

To address these research gaps, the proposed MEDISECURE: Secure Authentication and Authorization Framework for Healthcare IoT Systems integrates fingerprint-based biometric

authentication, real-time Pulse and ECG monitoring, and centralized patient record management into a single platform. The system combines an Arduino UNO R4 WiFi, R307/AS608 Fingerprint Sensor, Pulse Sensor, AD8232 ECG Sensor, React.js, Node.js with Express.js, and MySQL to provide secure authentication, live health monitoring, electronic prescription management, and role-based dashboards for doctors, patients, and hospital staff. This integrated approach enhances healthcare security, improves operational efficiency, and supports the implementation of modern smart hospital environments.

METHODOLOGY / REPORT ON PRESENT INVESTIGATION

3.1 Introduction

The proposed MEDISECURE: Secure Authentication and Authorization Framework for Healthcare IoT Systems is designed to provide a secure and efficient healthcare management platform by integrating biometric authentication, Internet of Things (IoT) devices, embedded systems, and web technologies. The system enables secure patient identification, real-time health monitoring, and centralized management of electronic medical records through an integrated hardware and software architecture.

The development of the system follows a modular approach, where each module performs a specific function while maintaining seamless communication with other components. The hardware section is responsible for patient authentication and acquisition of physiological data, whereas the software section manages user interaction, database operations, and communication between different modules. This integrated architecture ensures secure data transmission, efficient healthcare management, and reliable monitoring of patient health parameters.

The proposed methodology consists of patient registration, fingerprint-based authentication, real-time acquisition of Pulse and ECG signals, storage of patient information in a centralized MySQL database, and visualization of healthcare information through dedicated dashboards developed using React.js. Communication between the Arduino UNO R4 WiFi and the backend server is established through SerialPort, while REST APIs facilitate communication between the frontend and backend.

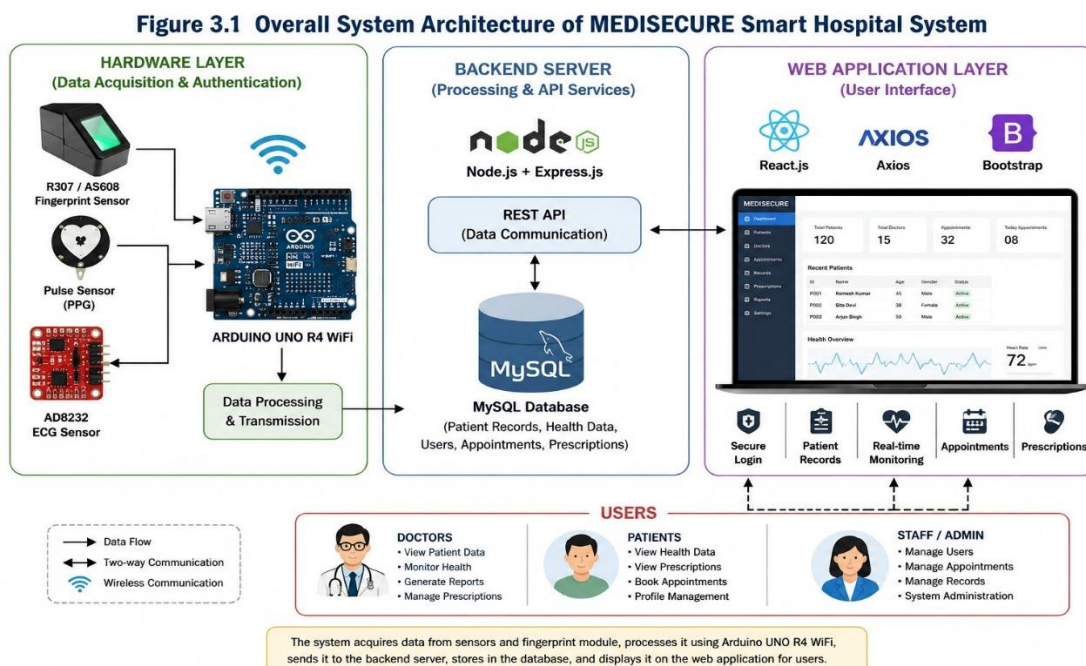


Figure 3.1 illustrates the overall architecture of the proposed MEDISECURE Smart Hospital System. The system consists of three major layers: the Hardware Layer, Backend Layer, and Web Application Layer.

The Hardware Layer includes the Arduino UNO R4 WiFi, R307/AS608 Fingerprint Sensor, Pulse Sensor, and AD8232 ECG Sensor. The fingerprint sensor is used for secure patient authentication, while the Pulse Sensor and AD8232 ECG Sensor acquire real-time physiological parameters. The Arduino UNO R4 WiFi collects and processes the sensor data before transmitting it to the backend server through SerialPort communication.

The Backend Layer is developed using Node.js and Express.js, which process incoming sensor data and provide RESTful APIs for communication with the frontend. Patient information, authentication details, prescriptions, and health records are securely stored in the MySQL database.

The Web Application Layer is developed using React.js, with Axios used for API communication between the frontend and backend. The application provides separate dashboards for doctors, patients, and hospital staff, enabling secure login, patient registration, real-time health monitoring, prescription management, and medical record access.

Overall, the architecture ensures secure communication between hardware and software components, enabling efficient healthcare management through biometric authentication, IoT-based health monitoring, and centralized data storage.

3.2 Hardware Components

The hardware components form the foundation of the proposed MEDISECURE Smart Hospital System. These components are responsible for secure patient authentication and real-time acquisition of physiological parameters. The system utilizes an Arduino UNO R4 WiFi as the central controller, interfaced with a R307/AS608 Fingerprint Sensor, Pulse Sensor, and AD8232 ECG Sensor. The collected data is transmitted to the backend server for processing, storage, and visualization through the web application.

3.2.1 Arduino UNO R4 WiFi

The Arduino UNO R4 WiFi serves as the central processing unit of the hardware system. It is responsible for acquiring data from the fingerprint sensor, pulse sensor, and ECG sensor. The board processes the collected data and communicates with the backend server through serial communication.

The Arduino UNO R4 WiFi was selected because of its improved processing capability, built-in WiFi support, sufficient GPIO pins, and compatibility with various healthcare sensors. It provides reliable communication between the embedded hardware and the software application.

Specifications:

- Microcontroller: Renesas RA4M1 (32-bit ARM Cortex-M4)
- Operating Voltage: 5 V
- Digital I/O Pins: 14
- Analog Input Pins: 6
- Communication: UART, SPI, I²C, USB
- Built-in WiFi Module

Role in the Project:

- Controls all hardware components.
- Collects pulse and ECG sensor data.
- Interfaces with the fingerprint sensor.
- Sends sensor readings to the backend through SerialPort communication.

Figure 3.2 Arduino UNO R4 WiFi



3.2.2 R307/AS608 Fingerprint Sensor

The R307/AS608 Fingerprint Sensor is used to provide secure biometric authentication for patients. It captures the patient's fingerprint, extracts unique fingerprint features, and stores them as digital templates within the sensor memory. During login, the captured fingerprint is compared with the stored templates to verify the patient's identity.

Fingerprint authentication eliminates the need for passwords and ensures that only authorized patients can access their medical records.

Specifications:

- Optical Fingerprint Recognition
- Operating Voltage: 3.6–6 V
- UART Serial Communication
- Internal Template Storage
- Fast Identification and Verification

Role in the Project:

- Patient registration.
- Fingerprint enrollment.
- Fingerprint verification.
- Secure patient authentication.

Figure 3.3 Fingerprint Sensor



3.2.3 Pulse Sensor

The Pulse Sensor is used to measure the patient's heart rate in real time. It detects changes in blood flow using the principle of photoplethysmography (PPG). The measured pulse signal is processed by the Arduino UNO R4 WiFi and transmitted to the backend for storage and visualization.

Continuous pulse monitoring assists healthcare professionals in assessing the patient's cardiovascular condition.

Specifications:

- Operating Voltage: 3.3–5 V
- Analog Output
- Optical PPG Sensor
- Low Power Consumption

Role in the Project:

- Measures heart rate.
- Sends pulse data to Arduino.
- Displays pulse readings on the patient dashboard.

Figure 3.4 Pulse Sensor



3.2.4 AD8232 ECG Sensor

The AD8232 ECG Sensor is designed to acquire electrocardiogram (ECG) signals from the patient. The sensor measures the electrical activity of the heart through surface electrodes attached to the patient's body. The ECG waveform is transmitted to the Arduino for processing and displayed in the healthcare application.

The AD8232 module provides reliable ECG acquisition while maintaining low power consumption, making it suitable for Healthcare IoT applications.

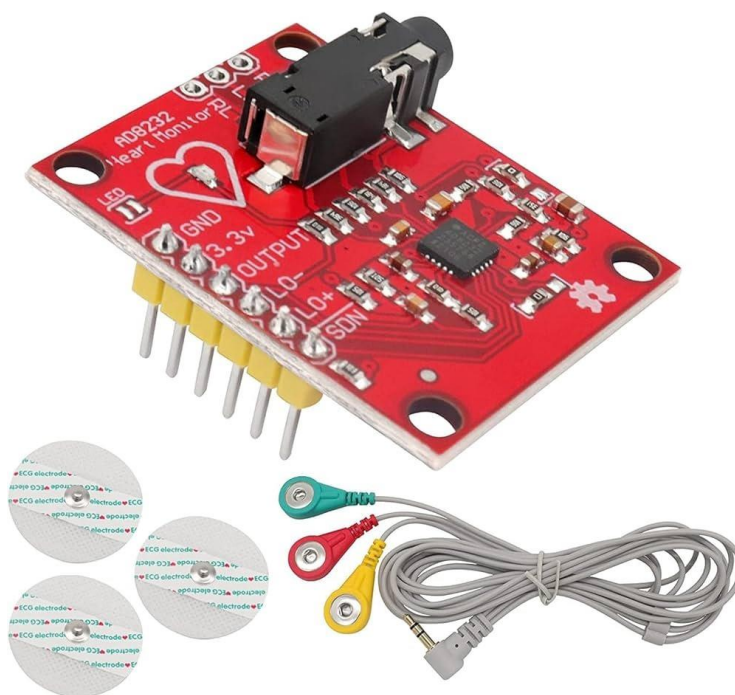
Specifications:

- Operating Voltage: 3.3–5 V
- Analog Output
- Three-Electrode Configuration
- Low Noise Signal Conditioning

Role in the Project:

- Captures ECG signals.
- Sends ECG data to Arduino.
- Enables real-time cardiac monitoring.
- Stores ECG readings in the database.

Figure 3.5 AD8232 ECG Sensor



Hardware Integration

All hardware components are connected to the Arduino UNO R4 WiFi, which acts as the central controller. The fingerprint sensor provides secure biometric authentication, while the Pulse Sensor and AD8232 ECG Sensor continuously monitor the patient's physiological parameters. The Arduino processes these inputs and transmits the collected data to the Node.js backend through SerialPort communication. The backend stores the information in the MySQL database and makes it available to the React-based web application for doctors, patients, and hospital staff.

3.4 System Workflow

The MEDISECURE Smart Hospital System follows a systematic workflow to ensure secure patient authentication, real-time health monitoring, and efficient medical record management. The workflow begins with patient registration, where personal information is entered into the system and a unique fingerprint is enrolled using the fingerprint sensor. The fingerprint template is securely stored and associated with the patient's record in the database.

During subsequent visits, the patient authenticates using the registered fingerprint. The fingerprint sensor verifies the identity, and upon successful authentication, the patient's dashboard is displayed. After authentication, the Pulse Sensor and AD8232 ECG Sensor acquire the patient's physiological parameters. The Arduino UNO R4 WiFi processes these readings and transmits the data to the Node.js backend through SerialPort communication.

The backend stores the received data in the MySQL database and provides it to the React.js frontend through REST APIs. Doctors can securely access patient records, review vital signs, and update diagnoses and prescriptions. Patients can view their medical history, prescriptions, and health reports through their dashboard. This workflow ensures secure access, efficient data management, and real-time monitoring within the healthcare system.

Figure 3.6 System Workflow of MEDISECURE Smart Hospital System

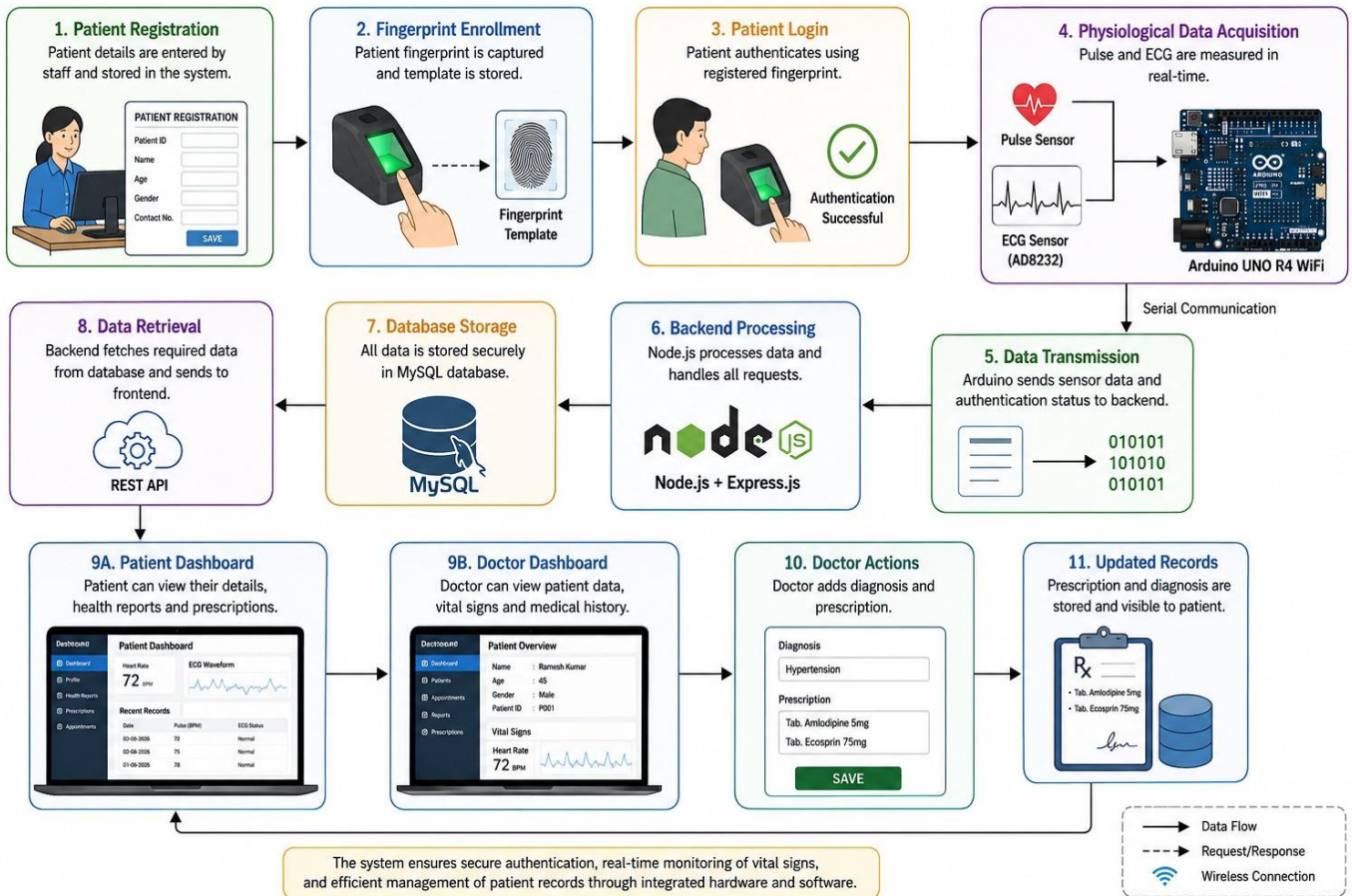


Figure 3.6 illustrates the complete workflow of the proposed MEDISECURE Smart Hospital System. The workflow begins with patient registration, where personal details are entered into the system and a unique fingerprint is enrolled using the fingerprint sensor. The enrolled fingerprint template is securely stored and linked to the patient's profile.

When the patient visits the hospital again, fingerprint authentication is performed to verify the patient's identity. Upon successful authentication, the Arduino UNO R4 WiFi collects physiological parameters from the Pulse Sensor and AD8232 ECG Sensor. The acquired data is transmitted to the Node.js backend through SerialPort communication.

The backend processes the received information and stores it in the MySQL database. Using REST APIs, the React.js frontend retrieves the stored data and displays it on the respective dashboards. Doctors can access patient details, monitor health parameters, and update diagnoses and prescriptions, while patients can securely view their health reports and medical history through their dashboard. This workflow ensures secure authentication, efficient healthcare management, and real-time monitoring of patient health.

3.5 Database Design

The MEDISECURE Smart Hospital System utilizes a MySQL relational database to securely store and manage all healthcare-related information. The database acts as the central repository for patient records, authentication details, prescriptions, and sensor readings. A structured database design ensures efficient storage, retrieval, and management of healthcare data while maintaining data consistency and integrity.

The database contains separate tables for patients, doctors, hospital staff, prescriptions, and sensor data. Each patient is assigned a unique Patient ID, which is associated with the corresponding fingerprint ID and medical records. The prescription table maintains diagnosis and medication details, while the sensor data table stores real-time pulse and ECG measurements. Relationships between these tables enable efficient retrieval of patient information and support seamless interaction between the frontend and backend.

The MySQL database communicates with the Node.js backend through SQL queries, while the backend exposes REST APIs for the React.js frontend. This centralized architecture ensures secure access to healthcare information and simplifies hospital record management.

Figure 3.7 Entity Relationship (ER) Diagram of the MEDISECURE Database

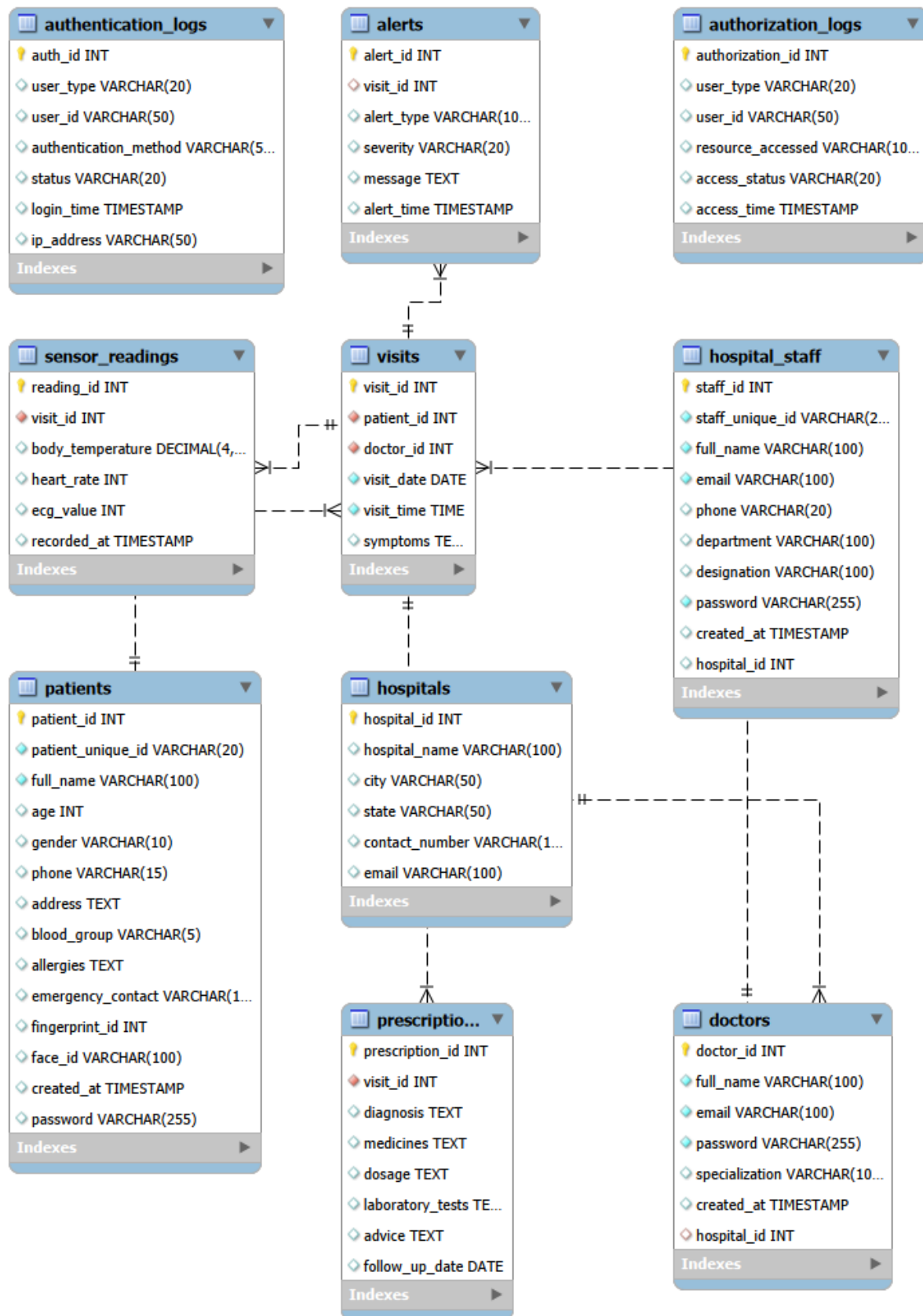


Figure 3.7 presents the Entity Relationship (ER) Diagram of the MEDISECURE Smart Hospital database. The database is designed using MySQL and consists of multiple interconnected tables that efficiently manage patient information, authentication details, hospital staff records, doctor information, sensor readings, prescriptions, hospital details, and system logs.

The patients table serves as the central repository for storing patient information such as patient ID, personal details, fingerprint ID, face ID (reserved for future use), contact information, and login credentials. Each patient record is associated with one or more visits, which store visit date, symptoms, and the assigned doctor.

The sensor_readings table stores physiological parameters including body temperature, heart rate, and ECG readings collected from the Arduino UNO R4 WiFi through the Pulse Sensor and AD8232 ECG Sensor. These readings are linked to the corresponding patient visit for maintaining complete medical records.

The prescriptions table maintains diagnosis details, prescribed medicines, dosage, laboratory tests, medical advice, and follow-up dates provided by doctors. The doctors and hospital_staff tables store authentication and profile information for authorized hospital personnel.

Additional tables such as authentication_logs, authorization_logs, and alerts are included to record login activities, access history, and system-generated alerts, thereby improving system security and auditability. The hospitals table maintains information about the healthcare institution associated with the records.

The relationships between these tables ensure data integrity, minimize redundancy, and support efficient retrieval of patient records. This relational database architecture enables secure storage, efficient management, and quick access to healthcare information, making it suitable for modern Healthcare IoT applications.

3.6 System Implementation

The MEDISECURE Smart Hospital System was developed by integrating embedded hardware, IoT sensors, a backend server, a relational database, and a web-based frontend into a unified healthcare management platform. The implementation follows a modular approach, where each module performs a specific function while communicating seamlessly with other components. This architecture ensures secure authentication, real-time monitoring of patient health, and efficient management of medical records.

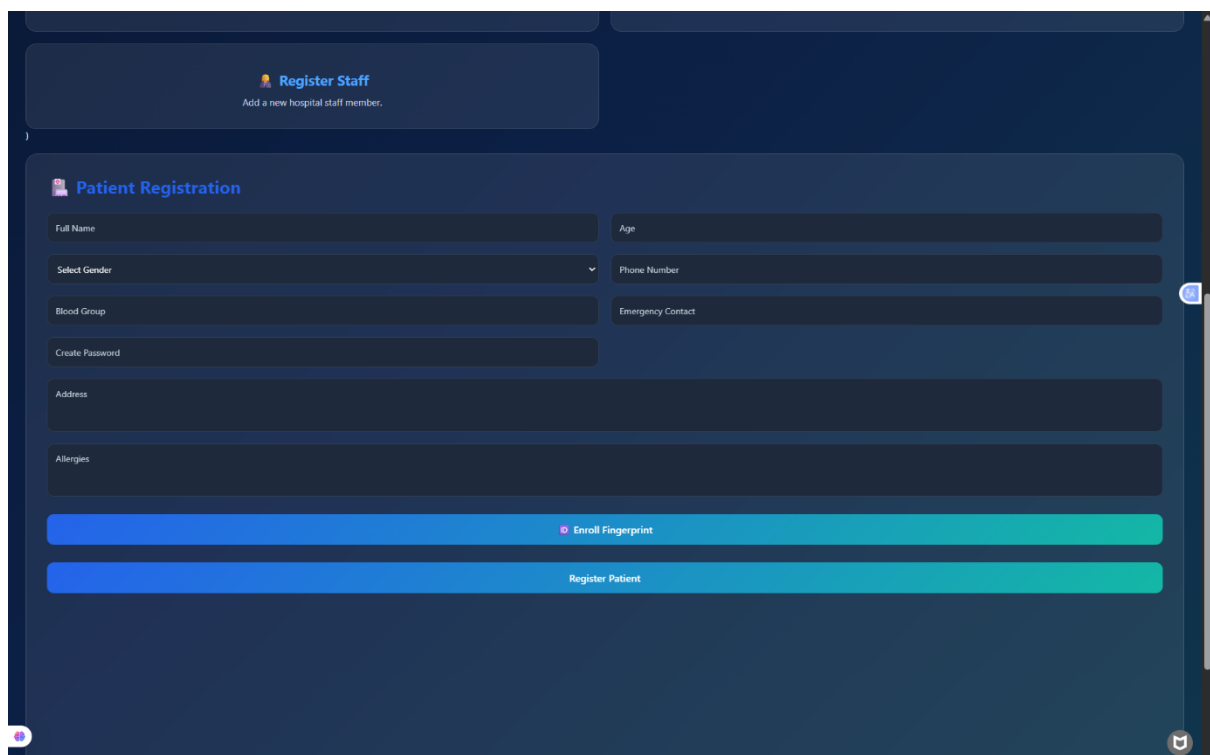
The implementation consists of the following modules:

3.6.1 Patient Registration Module

The patient registration module allows hospital staff to register new patients by entering their personal information such as Patient ID, name, age, gender, contact details, and other basic information. During registration, the patient's fingerprint is enrolled using the R307/AS608 Fingerprint Sensor. The generated fingerprint ID is associated with the patient's record and stored securely in the MySQL database.

This module ensures that every patient has a unique identity within the system, enabling secure authentication during future hospital visits.

Figure 3.8 Patient Registration Page



The screenshot displays the 'Patient Registration' interface. At the top, there is a 'Register Staff' button with the subtext 'Add a new hospital staff member.' Below this, the 'Patient Registration' section is highlighted. It contains the following fields: 'Full Name', 'Age', 'Select Gender' (a dropdown menu), 'Phone Number', 'Blood Group', 'Emergency Contact', 'Create Password', 'Address', and 'Allergies'. At the bottom of the form, there are two large, prominent buttons: 'Enroll Fingerprint' and 'Register Patient'.

3.6.2 Fingerprint Authentication Module

The fingerprint authentication module provides secure biometric verification of patients. When a patient places a finger on the fingerprint sensor, the Arduino UNO R4 WiFi captures the fingerprint image and compares it with the templates stored in the sensor memory. If a matching template is found, the corresponding patient information is retrieved from the database and the patient dashboard is displayed.

This biometric authentication mechanism improves security and eliminates the need for remembering passwords.

3.6.3 Pulse and ECG Monitoring Module

The Pulse Sensor and AD8232 ECG Sensor continuously acquire physiological parameters from the patient. The Arduino UNO R4 WiFi processes these analog signals and transmits the measured values to the Node.js backend through SerialPort communication. The backend stores the readings in the MySQL database and makes them available for visualization on the web application.

This module enables healthcare professionals to monitor patient health in real time and maintain a digital record of vital parameters.

Figure 3.9 Real-Time ECG Waveform Acquired Using the AD8232 ECG Sensor

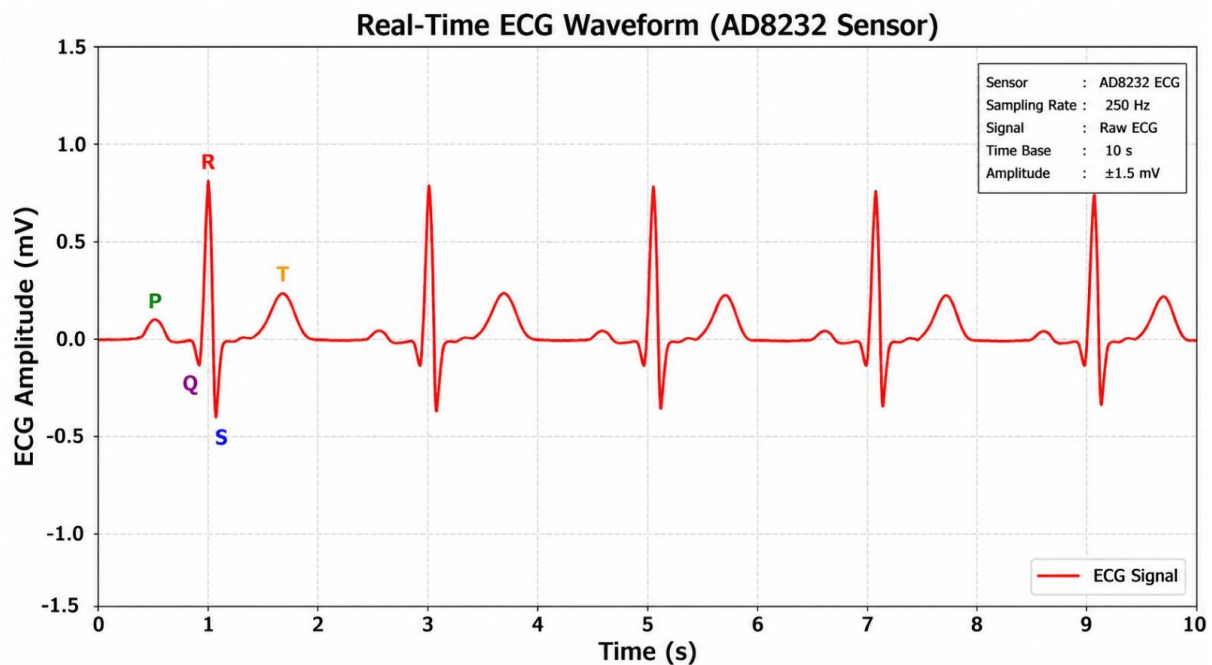


Figure 3.9 illustrates the real-time ECG waveform acquired using the AD8232 ECG sensor integrated with the Arduino UNO R4 WiFi. The ECG waveform represents the electrical activity of the heart and consists of the P wave, QRS complex, and T wave, which correspond to different stages of the cardiac cycle. These signals are captured using three ECG electrodes placed on the patient's body.

The Arduino UNO R4 WiFi continuously reads the analog ECG signal from the AD8232 sensor and transmits the acquired data to the Node.js backend through SerialPort communication. The backend processes the received data and stores it in the MySQL database, making it available for visualization on the React.js dashboard. Real-time ECG monitoring enables healthcare professionals to observe cardiac activity, detect abnormalities, and maintain digital health records for future clinical reference.

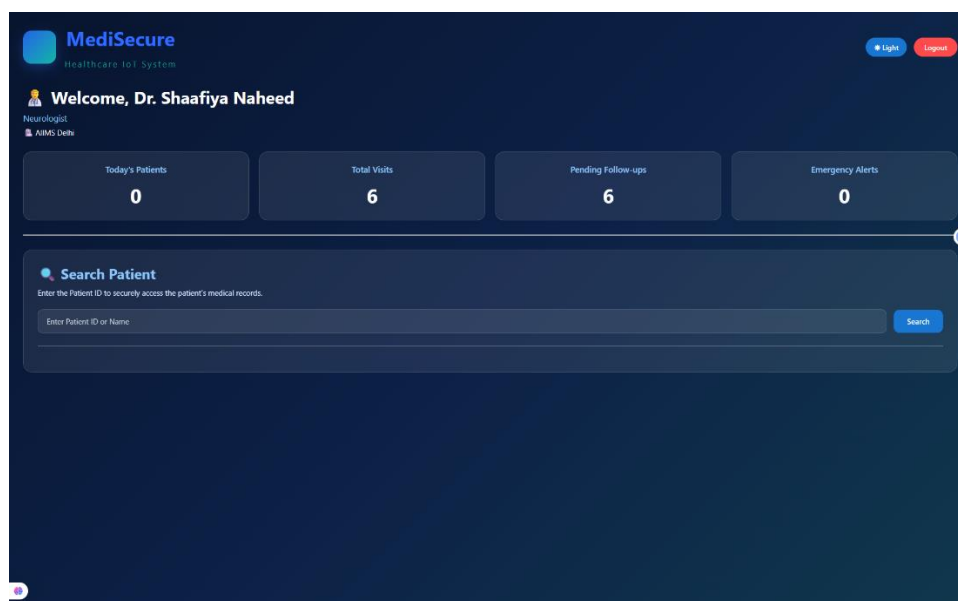
The integration of ECG monitoring with the MEDISECURE platform enhances patient care by providing continuous cardiac monitoring, secure data storage, and easy accessibility of health records through a centralized Healthcare IoT system.

3.6.4 Doctor Dashboard

The Doctor Dashboard provides authorized doctors with secure access to patient information, medical history, and real-time health parameters. After successful login, doctors can view patient details, monitor pulse rate and ECG readings, and add diagnoses and prescriptions. The updated medical information is stored in the MySQL database and becomes immediately available to the patient.

The dashboard provides a centralized interface for managing patient records, reducing paperwork and improving the efficiency of hospital operations.

Figure 3.10 Patient Dashboard



3.6.5 Patient Dashboard

The Patient Dashboard allows authenticated patients to securely access their personal information, health reports, prescriptions, and previous medical history. Real-time pulse and ECG readings are displayed through an easy-to-use graphical interface. Patients can review their prescriptions and monitor their health status without requiring direct access to the database.

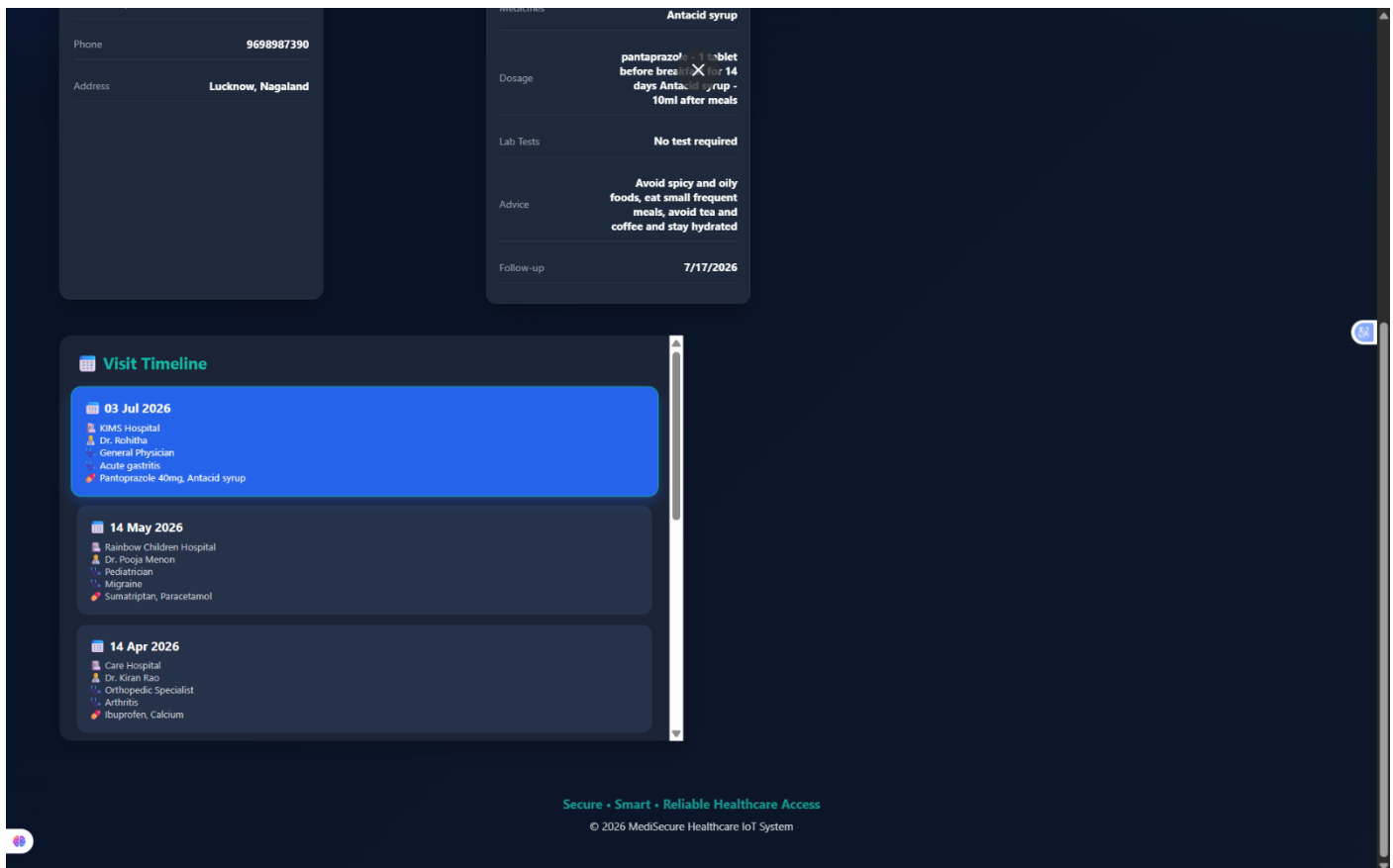
The dashboard improves transparency and enables patients to stay informed about their healthcare information.

Figure 3.11 Patient Dashboard

The screenshot displays the MediSecure Patient Dashboard. At the top left is the MediSecure logo with the tagline 'Healthcare IoT System'. A 'Light/Dark' theme toggle is in the top right. A 'PATIENT DASHBOARD' label is centered at the top. A 'Welcome, Balveer Bhavsar' message is prominently displayed, followed by the text 'Secure access to your healthcare information'. A 'Back' button is located on the left. The dashboard is divided into two main columns. The left column, titled 'Personal Information', contains fields for Patient ID (PAT0067), Name (Balveer Bhavsar), Age (85), Gender (Male), Blood Group (B+), Phone (9698987390), and Address (Lucknow, Nagaland). The right column, titled 'Selected Visit', displays details for a visit on 7/3/2026, including symptoms (Burning sensation in stomach after meals, nausea and acidity for 5 days), diagnosis (Acute gastritis), medicines (Pantoprazole 40mg, Antacid syrup), dosage (pantaprazole - 1 tablet before breakfast for 14 days, Antacid syrup - 10ml after meals), lab tests (No test required), advice (Avoid spicy and oily foods, eat small frequent meals, avoid tea and coffee and stay hydrated), and a follow-up date of 7/17/2026. A 'Visit Timeline' section is partially visible at the bottom left.

Personal Information	
Patient ID	PAT0067
Name	Balveer Bhavsar
Age	85
Gender	Male
Blood Group	B+
Phone	9698987390
Address	Lucknow, Nagaland

Selected Visit	
Visit Date	7/3/2026
Symptoms	Burning sensation in stomach after meals, nausea and acidity for 5 days
Diagnosis	Acute gastritis
Medicines	Pantoprazole 40mg, Antacid syrup
Dosage	pantaprazole - 1 tablet before breakfast for 14 days Antacid syrup - 10ml after meals
Lab Tests	No test required
Advice	Avoid spicy and oily foods, eat small frequent meals, avoid tea and coffee and stay hydrated
Follow-up	7/17/2026



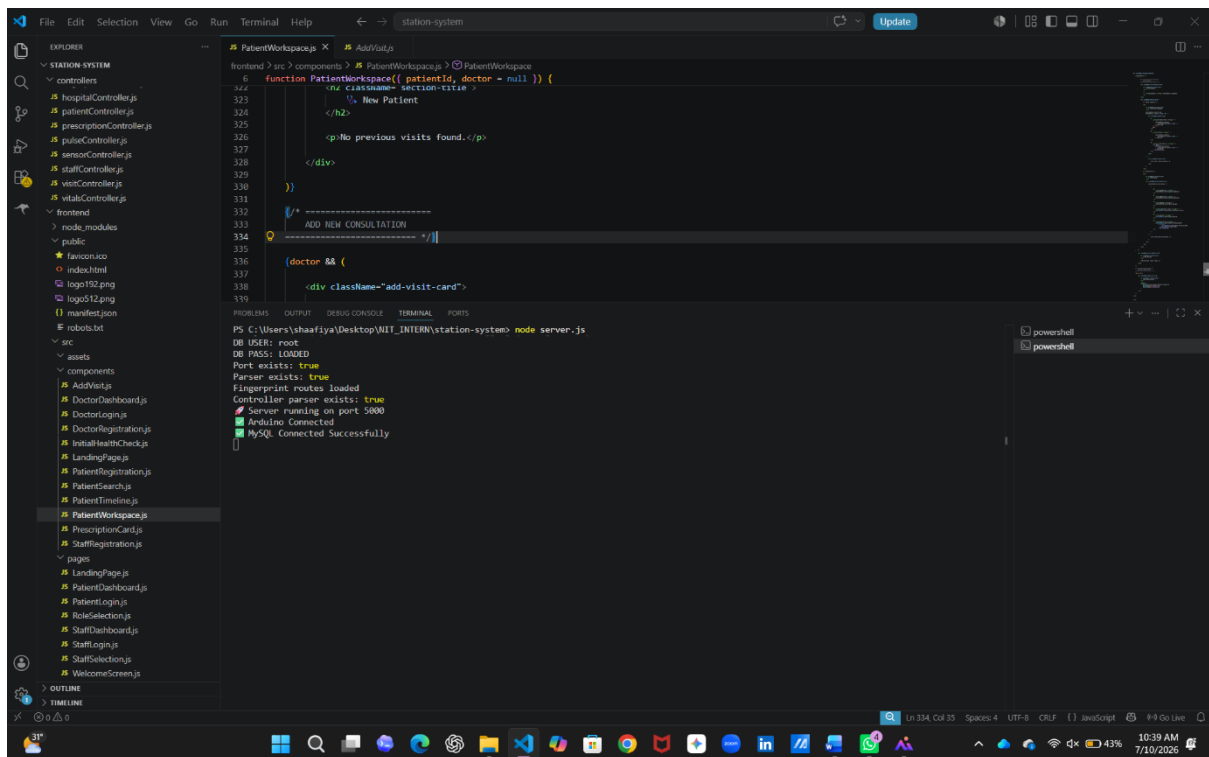
3.6.6 Backend and Database Integration

The backend of the MEDISECURE system is developed using Node.js and Express.js. It acts as an intermediary between the React.js frontend, Arduino hardware, and the MySQL database. REST APIs are implemented to handle patient registration, authentication, retrieval of medical records, storage of sensor readings, and prescription management.

SerialPort communication is used to establish a connection between the Arduino UNO R4 WiFi and the backend server. The backend receives fingerprint authentication status, pulse readings, and ECG data from the Arduino, processes the information, and stores it securely in the MySQL database.

This integrated architecture ensures efficient communication between all system components and enables real-time updating of healthcare information across the entire platform.

Figure 3.12 Backend Integration



RESULTS AND DISCUSSION

4.1 Introduction

The MEDISECURE Smart Hospital System was successfully developed and tested by integrating embedded hardware, IoT sensors, a web-based application, and a centralized database. The system was evaluated to verify secure patient authentication, real-time health monitoring, data transmission, database connectivity, and dashboard functionality. Experimental testing demonstrated successful communication between the Arduino UNO R4 WiFi, Node.js backend, MySQL database, and React.js frontend. This chapter presents the implementation results and discusses the overall performance of the developed system.

4.2 Patient Registration Result

The patient registration module was tested by registering multiple patients through the web application. The entered patient information was successfully validated and stored in the MySQL database. During registration, the patient's fingerprint ID was linked with the corresponding patient record, enabling secure biometric authentication for future logins.

The registration process was completed successfully without data loss, demonstrating reliable communication between the frontend, backend, and database.

4.3 Fingerprint Authentication Result

The fingerprint authentication module was tested using the R307/AS608 Fingerprint Sensor. Previously enrolled fingerprints were successfully recognized by the sensor, and the corresponding patient information was retrieved from the MySQL database. After successful authentication, the patient dashboard was displayed automatically.

The biometric authentication process eliminated the need for password-based login and ensured secure access to patient medical records.

4.4 Pulse and ECG Monitoring Result

The Pulse Sensor and AD8232 ECG Sensor successfully measured the patient's physiological parameters in real time. The Arduino UNO R4 WiFi acquired the analog sensor signals and transmitted them to the backend through SerialPort communication. The received data was stored in the MySQL database and displayed on the React.js dashboard.

The ECG waveform clearly represented the electrical activity of the heart, while the pulse sensor accurately measured heart rate. The obtained results demonstrated the capability of the proposed system to perform continuous health monitoring.

4.5 Doctor Dashboard Result

The Doctor Dashboard successfully displayed patient details, medical history, pulse readings, and ECG information. Doctors were able to update diagnoses and prescriptions, which were immediately stored in the MySQL database.

The dashboard provided a centralized interface for efficient patient management and reduced the dependency on paper-based records.

4.6 Patient Dashboard Result

The Patient Dashboard displayed patient information, prescriptions, medical history, and real-time health reports. Authenticated patients could securely access their records without requiring manual assistance from hospital staff.

The dashboard improved accessibility and transparency of healthcare information.

4.7 Database Result

The MySQL database successfully stored patient records, fingerprint IDs, prescriptions, doctor information, and sensor readings. All CRUD (Create, Read, Update, Delete) operations were executed correctly through the Node.js backend.

The database demonstrated reliable storage and retrieval of healthcare information while maintaining data consistency

4.8 Discussion

The experimental results demonstrate that the MEDISECURE Smart Hospital System successfully integrates biometric authentication, real-time health monitoring, and centralized medical record management into a single Healthcare IoT platform. The system provides secure fingerprint-based authentication, continuous pulse and ECG monitoring, efficient database management, and intuitive dashboards for doctors and patients.

The integration of React.js, Node.js, Express.js, MySQL, and Arduino UNO R4 WiFi ensured smooth communication between hardware and software components. The developed system reduced manual documentation, improved healthcare security, and enabled efficient access to patient information. Overall, the implementation validates the feasibility of the proposed framework for modern smart healthcare applications.

Figure 4.1 Experimental Hardware Prototype of the MEDISECURE Smart Hospital System

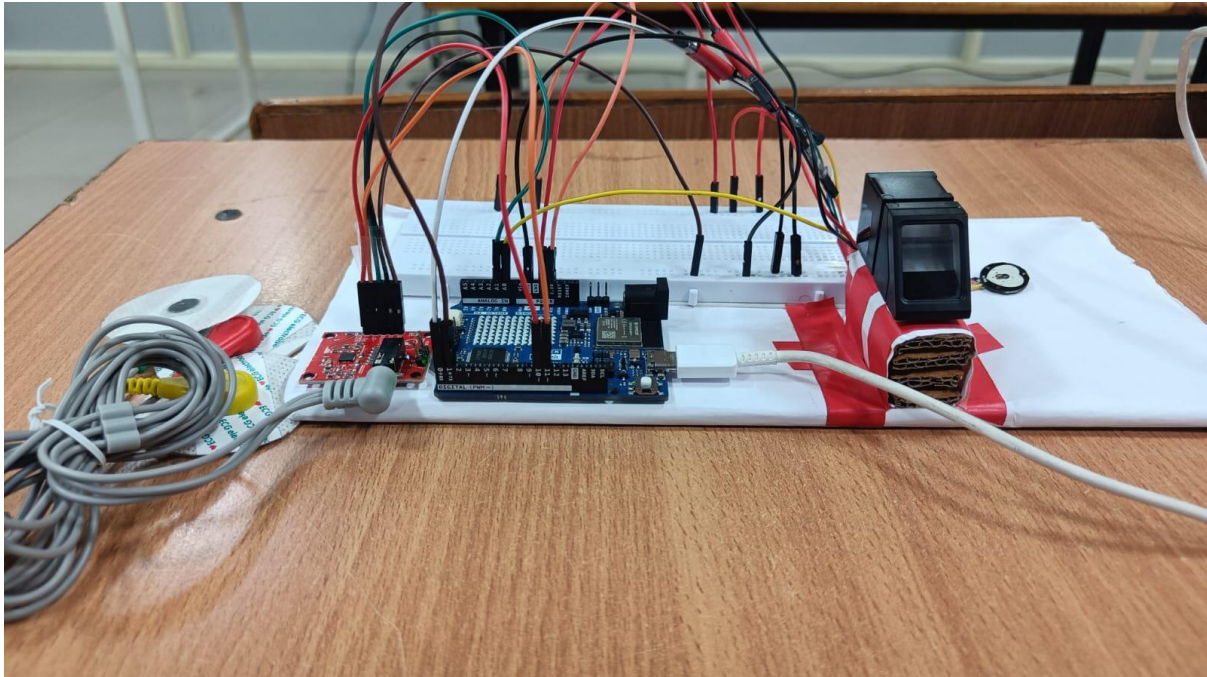


Figure 4.1 presents the experimental hardware prototype of the proposed MEDISECURE Smart Hospital System developed during the internship. The prototype consists of an Arduino UNO R4 WiFi, R307/AS608 Fingerprint Sensor, AD8232 ECG Sensor, ECG electrodes, breadboard, and the required interconnecting wires. The Arduino UNO R4 WiFi functions as the central controller, coordinating communication between the biometric sensor, physiological sensors, and the backend application.

The hardware prototype was successfully integrated with the React.js frontend, Node.js and Express.js backend, and MySQL database. During experimental testing, the system authenticated patients using fingerprint recognition, acquired real-time ECG and pulse signals, transmitted sensor data through SerialPort communication, and displayed the processed information on the web-based dashboard. The successful operation of the prototype demonstrates the effectiveness of the proposed MEDISECURE framework for secure authentication, real-time health monitoring, and digital healthcare management.

SUMMARY AND CONCLUSIONS

5.1 Summary

The MEDISECURE: Secure Authentication and Authorization Framework for Healthcare IoT Systems was successfully designed and implemented to provide a secure, reliable, and efficient smart healthcare management solution. The project integrates biometric authentication, embedded systems, IoT sensors, and modern web technologies to enhance patient identification and healthcare data management.

The developed system utilizes an Arduino UNO R4 WiFi as the central controller, interfaced with an R307/AS608 Fingerprint Sensor, Pulse Sensor, and AD8232 ECG Sensor for secure authentication and real-time health monitoring. The software architecture was developed using React.js for the frontend, Node.js with Express.js for the backend, and MySQL for centralized database management. Communication between the hardware and backend was established using the SerialPort library, while REST APIs enabled seamless interaction between the frontend and backend.

The system successfully performs patient registration, fingerprint-based authentication, real-time pulse and ECG monitoring, electronic medical record management, and prescription management through dedicated dashboards for patients and doctors. Experimental testing confirmed reliable communication between the hardware and software components, secure storage of healthcare data, and accurate visualization of patient information. The developed prototype demonstrates the practical application of Healthcare IoT technologies for improving hospital efficiency and enhancing patient care.

5.2 Conclusions

The proposed MEDISECURE Smart Hospital System successfully achieved the objectives of providing secure patient authentication and real-time healthcare monitoring within an integrated Healthcare IoT platform. The combination of fingerprint-based biometric authentication with continuous pulse and ECG monitoring ensures improved security, efficient patient identification, and accurate collection of physiological data.

The integration of Arduino UNO R4 WiFi, React.js, Node.js, Express.js, MySQL, and SerialPort communication resulted in a reliable and scalable system capable of supporting modern healthcare environments. The developed solution reduces manual record management, improves accessibility of medical information, and enhances communication between patients and healthcare professionals.

Overall, the project demonstrates that the integration of IoT devices, embedded systems, and web technologies can significantly improve the efficiency, security, and quality of healthcare services. The MEDISECURE framework provides a strong foundation for future smart hospital applications and contributes towards the digital transformation of healthcare systems.

5.3 Future Scope

The proposed MEDISECURE system can be further enhanced by incorporating additional features to improve functionality and scalability. Future improvements include:

- Integration of cloud computing for remote access to electronic health records.
- Development of a mobile application for patients and healthcare professionals.
- Implementation of AI and machine learning algorithms for disease prediction and health analysis.
- Addition of more IoT sensors such as blood pressure, SpO₂, blood glucose, and body temperature sensors.
- Secure cloud backup and data encryption for enhanced privacy and security.
- Integration with hospital information systems and telemedicine platforms for remote healthcare services.

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